Zi En Tham (Zen)

+61450190503 | thamzien@gmail.com | Github | Tableau Profile

Education

Bachelor of Computing Science (Honours) **August 2021 - August 2024**
University of Technology Sydney (UTS)

- Major: Artificial Intelligence and Data Analytics (WAM 86.58)

Skills

Programming language: SQL, Python, HTML/CSS, Java, PHP, Javascript, C++
Languages: English, Chinese (Native), Cantonese, Malay
Tools / Tools: Git, Node, React, TypeScript, TailWindCSS, Yi Framework, Laravel, Excel, Tableau, PowerBI

Work Experience

Web Developer **May 2025 – Present**
Trivesta, Sydney, NSW, Australia

- Built and maintained full-stack features for a corporate website and internal CRM.
- Built n8n automation workflows to handle client calls and inquiries, reducing manual processing time.
- Built predictive AI/ML churn models achieving 80%+ accuracy, and directly informing client retention strategies.
- Tools: Wordpress, Typescript, React, Vite, PHP, Javascript, Redux, Axios, MySQL, AntDesign, TailwindCSS, n8n, PowerBI

Operation Developer **September 2024 - Present**
Flash Media Solutions, Rydalmere, NSW, Australia

- Developed a modern e-commerce website serving over 20 products and 50 monthly visitors, migrating from a legacy stack to improve performance and maintainability
- Built an automated Excel stock and invoice tracking system that eliminated duplicate records across 250 inventory items.
- Designed Tableau dashboards visualizing sales trends, regional client distribution, and stock velocity.
- Managed client relationships and translated business requirements into technical solutions.
- Tools: PHP, JavaScript, GSAP, WordPress, Excel, Tableau

Teaching Assistant **January 2025 - July 2025**
University of Technology Sydney, Sydney, NSW, Australia

- Tutored and graded 40 students with written feedback on their research project
- Supported students applying Python ML libraries to academic projects
- Tools: Python, Scikit-learn, TensorFlow, LaTeX, Excel, Microsoft Word

Research Assistant **August 2024 - December 2024**
La Trobe University, Melbourne, Victoria, Australia

- Developed co-learning MedNext based model for brain haemorrhage detection, achieving 86% accuracy in classifying brain hemorrhage in CT scans.
- Adapted Bal-CE loss and post-hoc logit adjustment to improve class balance and achieved up to 21% improvement in accuracy and 20% in ROC AUC over previous state-of-the-art methods.
- Tools: Python, Docker, Scikit-learn, MedNext, CT Imaging

U: PASS Data Team **January 2023 - June 2024**
University of Technology Sydney, Sydney, NSW, Australia

- Collected, cleaned, and maintained attendance records for over 200 sessions, ensuring data integrity for academic performance reporting.

- Generated Excel reports and visualizations tracking study pattern trends and grade correlations, used by supervisor for program evaluation.
- Tools: Microsoft Excel

U: PASS Leader **January 2023 - June 2024**
University of Technology Sydney, Sydney, NSW, Australia

- Facilitated weekly peer-assisted study sessions for 25 computing science students, covering data structures and algorithms
- Prepared structured learning materials and exercises tailored to identified performance gaps, contributing to over 90% student satisfaction score outcome
- Reported student engagement metrics and session outcomes to faculty coordinators, informing curriculum support decisions for programming subjects
- Tools: Java, C++, Excel, Microsoft Word

Student Ambassador **July 2022 - December 2024**
University of Technology Sydney, Sydney, NSW, Australia

- Represented the university across 3 open days and over 10 events.
- Led 4 workshops covering course pathways, computing programs, and student life for audiences of 20 attendees
- Managed prospective student records and communication follow-ups using Salesforce CRM
- Tools: Tools: Salesforce, Python, Excel, Public Speaking

Technical Support Engineer Intern **January 2020 – Mar 2020**
Net Onboard Sdn Bhd, KL, Malaysia

- Created and organized network cables
- Prepared and deployed servers requested by clients
- Troubleshoot server issue reported by clients
- Deployed testing environment
- Planned and created the network architectures
- Recorded and documented meeting minutes for the technical support department
- Tools: Powershell Script, Linux, Ubuntu, CentOS, Virtual Machines

Projects

WEB APPLICATION

2D Drawing to 3D Model Web Tool **Ongoing 2026**

- Built browser-based tool converting 2D canvas drawings into exportable 3D GLB models
- Implemented Three.js extrusion pipeline with direct compatibility for major game engines
- Designed for mobile and desktop with future skeletal rigging support planned (ongoing)
- Tools: Three.js, JavaScript, WebGL, GLB Export

AI Study Assistant **Ongoing 2026**

- Built LLM-powered study platform with PDF upload and personalized concept explanation
- Implemented automated flashcard generation with accurate citation tracking per source
- Centralized fragmented study workflow previously spread across multiple AI chat platforms
- Tools: React, TypeScript, NVIDIA NIM API, PDF Processing, LangChain

Flash Media Website **Ongoing 2026**

- Migrated legacy Magento 1.0 e-commerce site to a maintainable WordPress stack
- Extended Elementor with custom GSAP animations to achieve a premium visual feel
- Delivered CMS architecture independently manageable by non-technical staff
- Tools: WordPress, Elementor, GSAP, HTML, CSS

Trivesta Website **December 2025**

- Built custom WordPress theme from scratch with Tailwind CSS and GSAP animations
- Configured full SEO setup including analytics, sitemap, and canonical links
- Established GitHub Actions CI/CD pipeline for fully automated deployments
- Tools: HTML, Tailwind CSS, JavaScript, WordPress, Figma, GitHub Actions

Gamify Physiotherapy (UNSW Hackathon 2025) **November 2025**

- Built a web app helping patients track and monitor knee rehabilitation exercises
- Integrated MediaPipe pose analysis for real-time exercise form validation feedback
- Developed full-stack solution within a hackathon timeframe at UNSW Kensington
- Tools: Next.js, TypeScript, TailwindCSS, Shadcn UI, Prisma ORM, PostgreSQL, Python, MediaPipe

Online Leave Management System **July 2023**

- Built full-stack Laravel application replacing a manual student leave request process
- Implemented role-based access control for student applicants and lecturer approvers
- Automated approval workflow reducing leave processing to a same-session turnaround
- Tools: Laravel, PHP, MySQL, Bootstrap, Blade

DATA ANALYTICS / DASHBOARDS

Anatomy of Kickstarter **February 2026**

- Analysed 200,000 Kickstarter campaigns spanning a full decade (2009–2020)
- Filtered advertising-disguised campaigns to surface genuine fundraising patterns and trends
- Visualized platform genre evolution from creative category dominance to tech-games parity
- Tools: Tableau, Data Analysis, Data Cleaning

Google Ads Performance Dashboard **February 2026**

- Built real-time Looker Studio dashboard integrating live Google Ads API data
- Configured threshold-based alerts and custom metric views
- Drove 30% conversion rate improvement through continuous data-driven campaign optimization

- Tools: Looker Studio, Google Ads API, Google Analytics

Sales Insights Dashboard **November 2025**
- Built Tableau dashboard visualizing sales performance, regional trends, and stock velocity
- Enabled sale representative to independently access business insights
- Tools: Tableau, Excel, Data Visualization

Inventory Management System **October 2025**
- Designed normalized Excel VBA database to replace a broken legacy CSV system
- Automated stock validation, warranty tracking, and aging inventory alert generation
- Eliminated duplicate records across 30 product SKUs entirely
- Tools: Excel VBA, Database Design, CSV Processing

Warranty Management System **September 2025**
- Automated warranty card generation using Excel VBA
- Integrated warranty tracking module directly with existing inventory database records
- Eliminated manual editing errors and provided a full client audit trail
- Tools: Excel VBA, Database Design, Document Generation

Australia Homelessness Analysis **April 2025**
- Analysed multi-source datasets revealing Victoria's disproportionately high homelessness rate
- Identified mental health as the primary driver among the 35–44 age group
- Published interactive Tableau dashboard enabling public exploration of key findings
- Tools: Tableau, Data Analysis, Public Data

Australia Economic Trade Analysis **September 2023**
- Analyzed 34 years of Australian international trade data using Tableau visualisations
- Focused on machinery and crude materials to surface meaningful long-term trade trends
- Delivered stakeholder-ready insights report for economic and policy decision-makers
- Tools: Tableau, Trade Data Analysis, Economic Research

Property Analysis in Mortdale **August 2023**
- Cleaned and analyzed ABS housing data to examine Mortdale suburb price trends
- Correlated property values with key demographic and community characteristic variables
- Produced actionable buyer and renter recommendations report for academic submission
- Tools: Excel, Data Analysis, ABS Data

AI MODEL / PREDICTIONS

Urban Heat Index Prediction for Clinical Laboratory (EY Open Science 2025) **March 2025**
- Built a Random Forest model predicting Urban Heat Index in Manhattan and Bronx
- Achieved 97% R^2 score through manual sourcing and integration of diverse datasets
- Extracted, mapped, and processed geospatial, time series, and numerical data formats
- Tools: Python, Random Forest, GeoPandas, Time Series Analysis

Anomaly Detection with LSTM for Clinical Laboratory (Honours Thesis) **June 2024**
- Built unsupervised LSTM system for real-time anomaly detection in clinical lab data
- Combined statistical QC methods with deep learning to enable proactive error detection
- Achieved 5-10% improvement over traditional QC baselines
- Tools: Python, LSTM, Neural Networks, Statistical Methods

Landmark Recognition for Vivid Sydney 2023 **July 2023**

- Trained ResNet50 CNN to classify over 10 VIVID Sydney art installations
- Achieved over 85% classification accuracy under variable real-world lighting conditions
- Deployed Android app with Firebase delivering real-time artist information on recognition
- Tools: Python, ResNet50, CNN, Android Studio, Java, Firebase

Referees

Dr Xu Wang
Senior Lecturer at School of Electrical and Data Engineering UTS
[Contact details available on request]

Eugene Nguyen
Formal Colleague at Trivesta
[Contact details available on request]